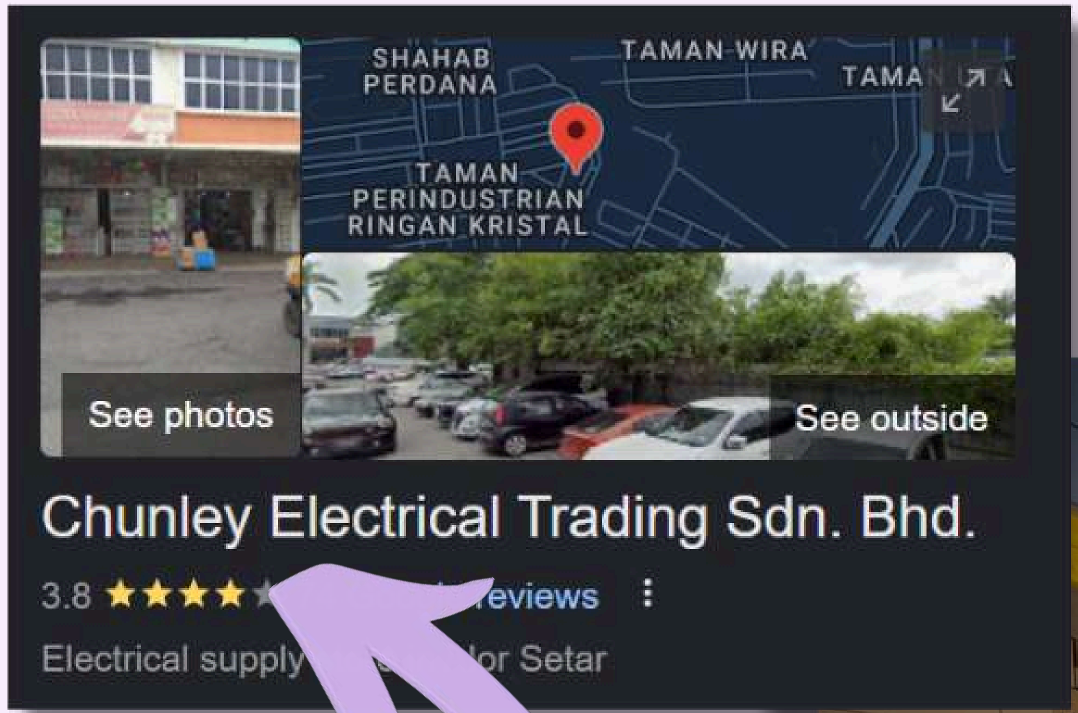
Our Team :

CISN

Speaker: TEOH JUN HONG

WHERE OUR IDEA COME FROM



TARGET USERS :
SMALL TO MEDIUM ELECTRICAL TRADING WAREHOUSE

PROBLEM

In SME warehouses, the moment a delivery truck arrives, a worker has to answer three questions at once:

1. **Is any of this stock already promised to a customer order today?** — if yes, it should not be binned at all.
2. **Where is the best bin to put the rest?** — depends on product weight, volume, sales velocity, customer base, blocked areas, and remaining bin capacity.
3. **What if something goes wrong?** — a forklift fails, a spill happens, an aisle is blocked.

Today this is solved by walking around, asking the manager, guessing, or memorising.
The result: **wasted time, wasted space, miscommunication, and unsafe routing.** A worker who is new to the floor cannot make these decisions at all.
Zai is built so that any worker — new or experienced — can make the optimal placement decision in seconds, just by talking to the system.

DOMAIN 1: AGENTIC WORKFLOW

INTELLIGENT WAREHOUSE RESTOCK MANAGER

Intelligent system which help organisation to optimize time, cost and space in warehouse

Operate 24/7

100% work

Cutting-edge technology

Easily access using phone

SYSTEM V.S

TRADITIONAL SYSTEM

Workers walk around to find space for stock

Waste time and may be not accurate

Workers make sure this product is not ordered by customers and need to arrange

Manager has the need to make sure this spot is the best for this specific stock

Miscommunication

If workers are new and not familar with the product they need to reach out manager

When accident happen they need to contact manag

Login system to record the location of stock

OUR INTELLIGENT SYSTEM

Workers insert the product_id and quantity arrived or describe it to AI

Accurate in short time

AI automatically check&update the product location to database after workers confirm

Workers trigger Accident Report Function when they describe accident happens to AI

MAIN FUNTION

CHAT

- Just describe your needs about restock and AI will asist you
- Feel free to ask question such as:

  what is K15VC

  is there any K15YC in our warehouse?

  A truck with 30 SCH-E8331 arrived where can I put it

CUSTOMER ORDER

- Workers no need to arrange the prouct which ordered by customers that day
- Our AI auto realise when workers try to arrange them into bins and immediately inform them only put the remaining to bin
- Our AI can read local Customer Order file in csv type by itself

ACCIDENT REPORT SYSTEM

- Workers just describe what they see and AI help them to :
  - - Report the case to manager
  - Block the bins in the Aisle that had been affected (wont suggest workers the affected bin before any update)
- Just send clear and AI will inform manager and unblock bins

DIRECTORY MAP

- Guide worker to find exact location of bins
- Visualize the bins status and whats the product inside
- Save workers time

TECHNOLOGY

AI AGENT

1. Visit the database (located in different department) to get the information it need [need tools to visit and update database]
2. By using tools and information it gets,find the optimal bin (slot) for specific product
3. Report the accident to manager via telegram and block the affected bins at the same time and update manager when it is clear
4. Check the local csv file (about customer order)
5. Inform the manager when customer orders arrived

USING TOOLS

AVOID HALLUCINATION

cool-eclair-a21518.netlify.app

OP: A truck of K15VC arrived

AI: How many units of 60 inch 3-Blade Ceiling Fan arrived?

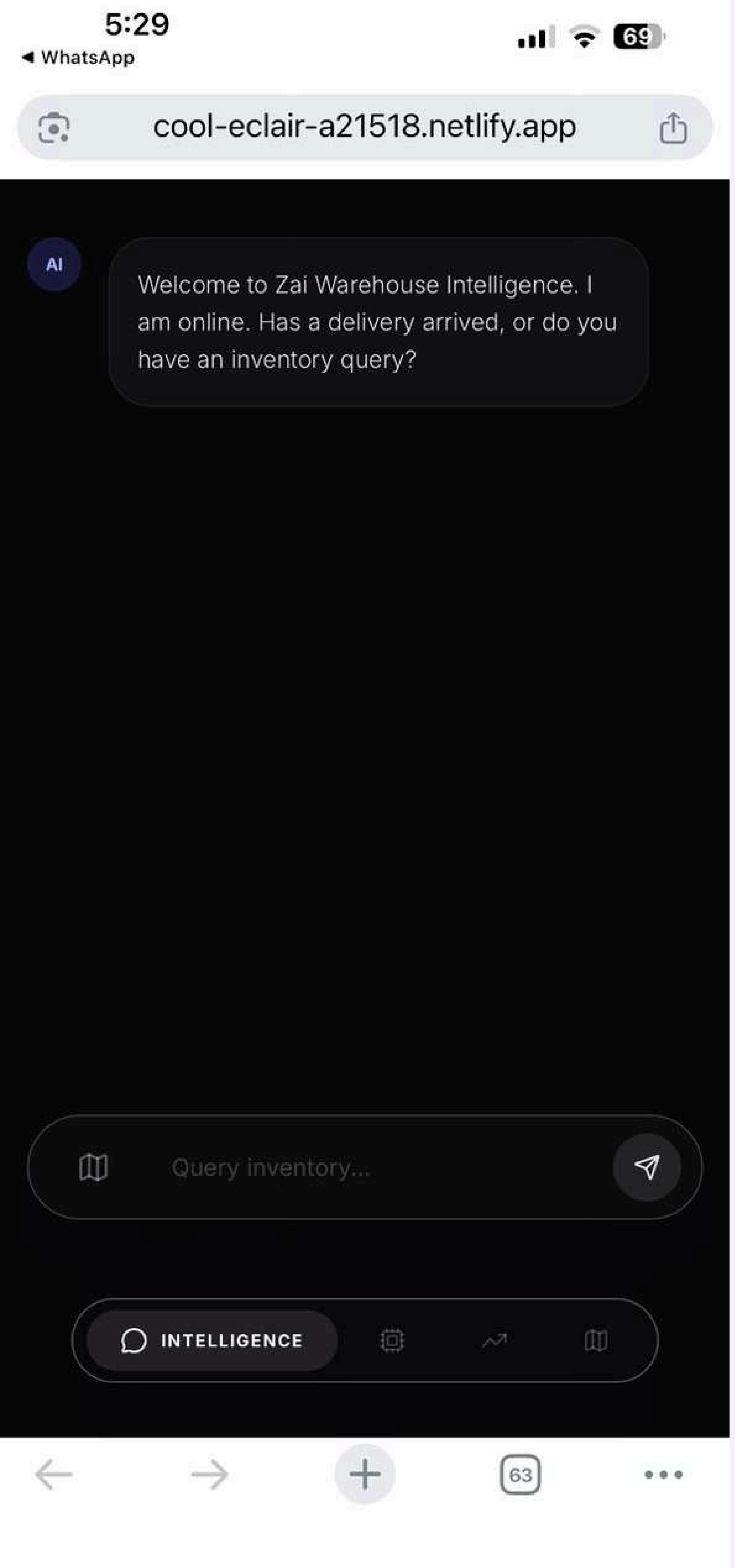
OP: 60

AI: Send 40 units to the Packing Counter for KL Rapid Construction (High priority). Bin A1-S1-L2-B1: place 18 units. Bin A1-S1-L2-B2: place 2 units. Reply "done" when you have placed the items.

OP: done

AI: Delivery of 60 units of K15VC is now complete. 40 units were sent to the Packing Counter for KL Rapid Construction, and 20 units were binned in A1-S1-L2-B1 and A1-S1-L2-B2. All bin updates have been committed.

DEMO


5:29
◀ WhatsApp
69
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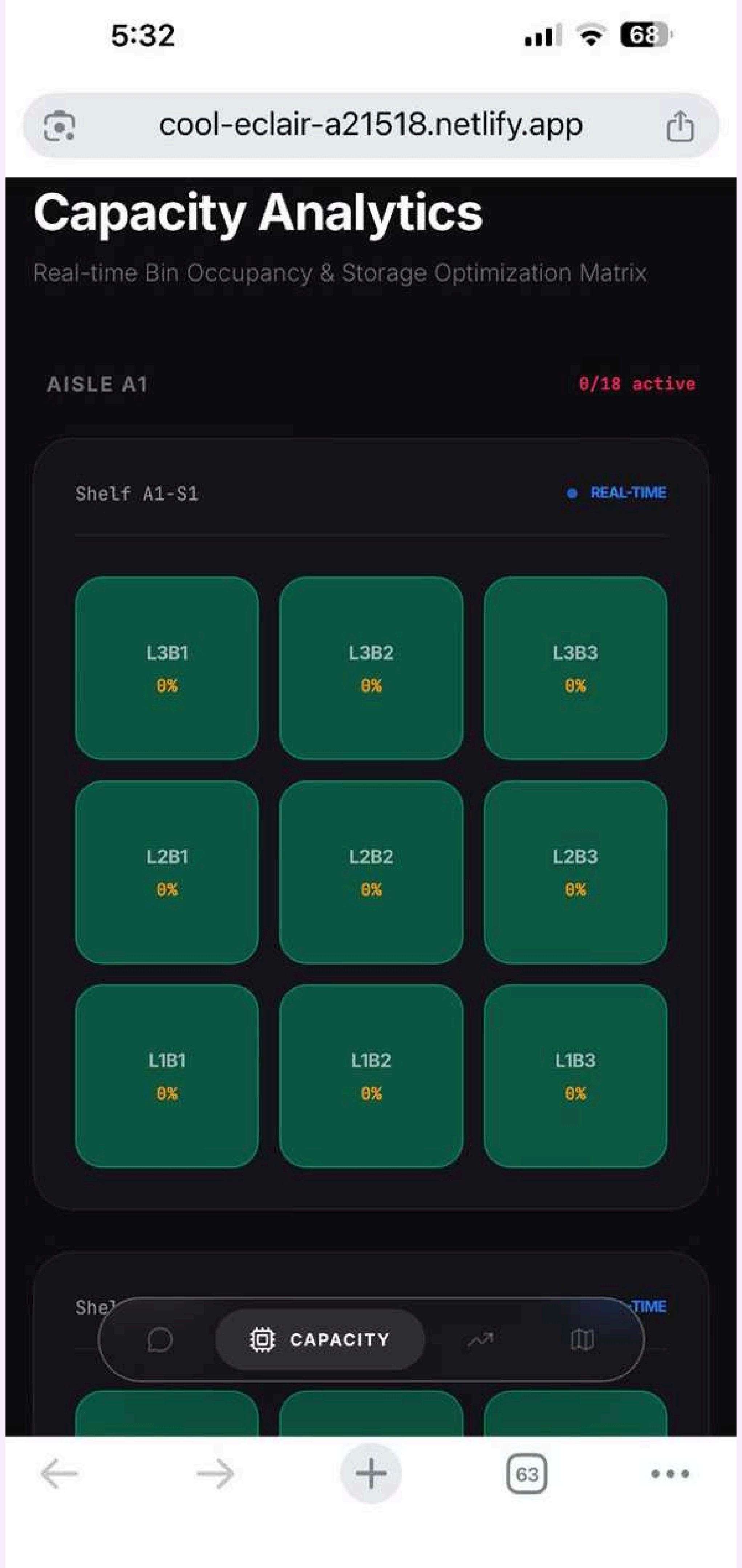
AI
Welcome to Zai Warehouse Intelligence. I am online. Has a delivery arrived, or do you have an inventory query?
Query inventory...
INTELLIGENCE
63


5:32
68
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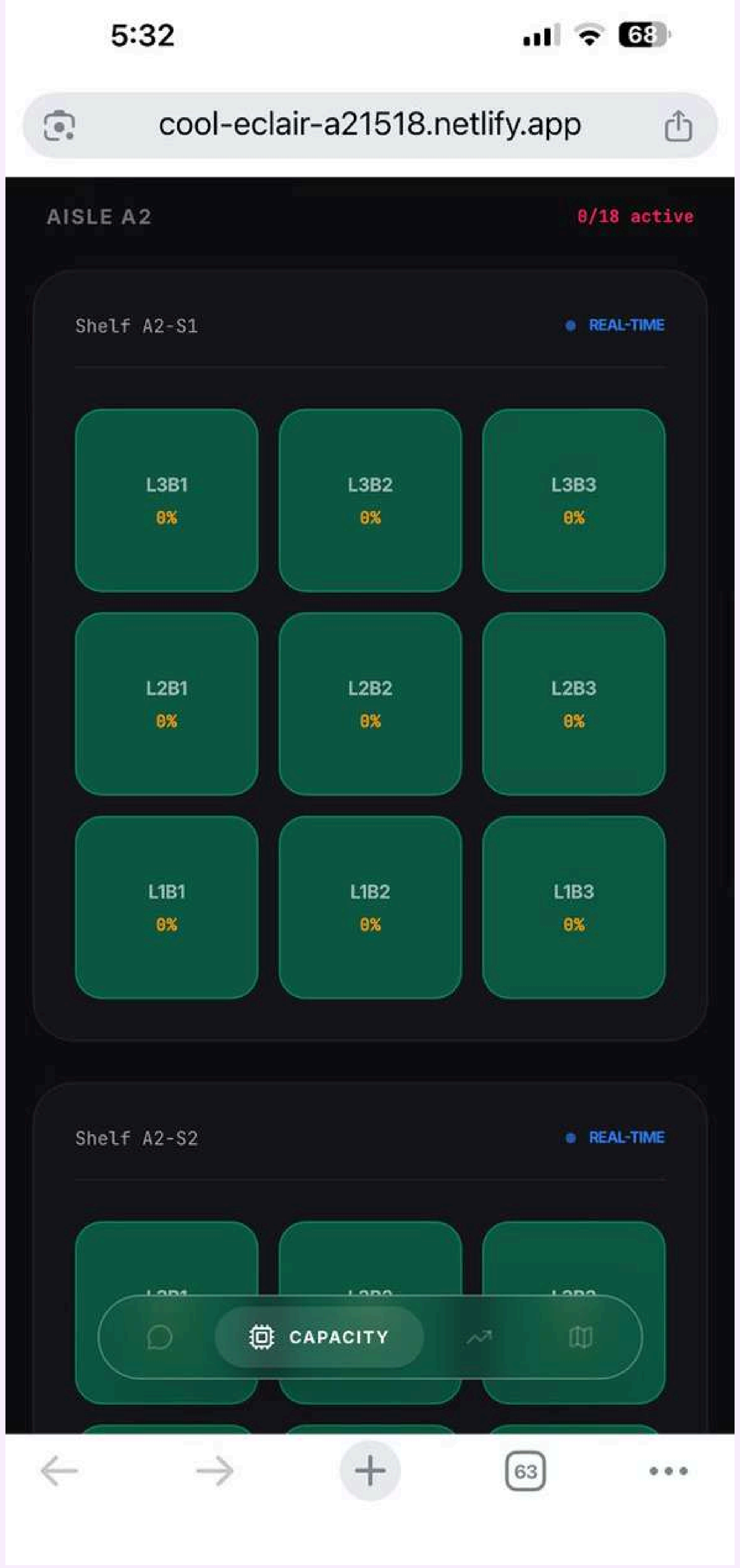
Capacity Analytics
Real-time Bin Occupancy & Storage Optimization Matrix
AISLE A1
0/18 active
Shelf A1-S1
REAL-TIME
L3B1
0%
L3B2
0%
L3B3
0%
L2B1
0%
L2B2
0%
L2B3
0%
L1B1
0%
L1B2
0%
L1B3
0%
CAPACITY
63


5:32
68
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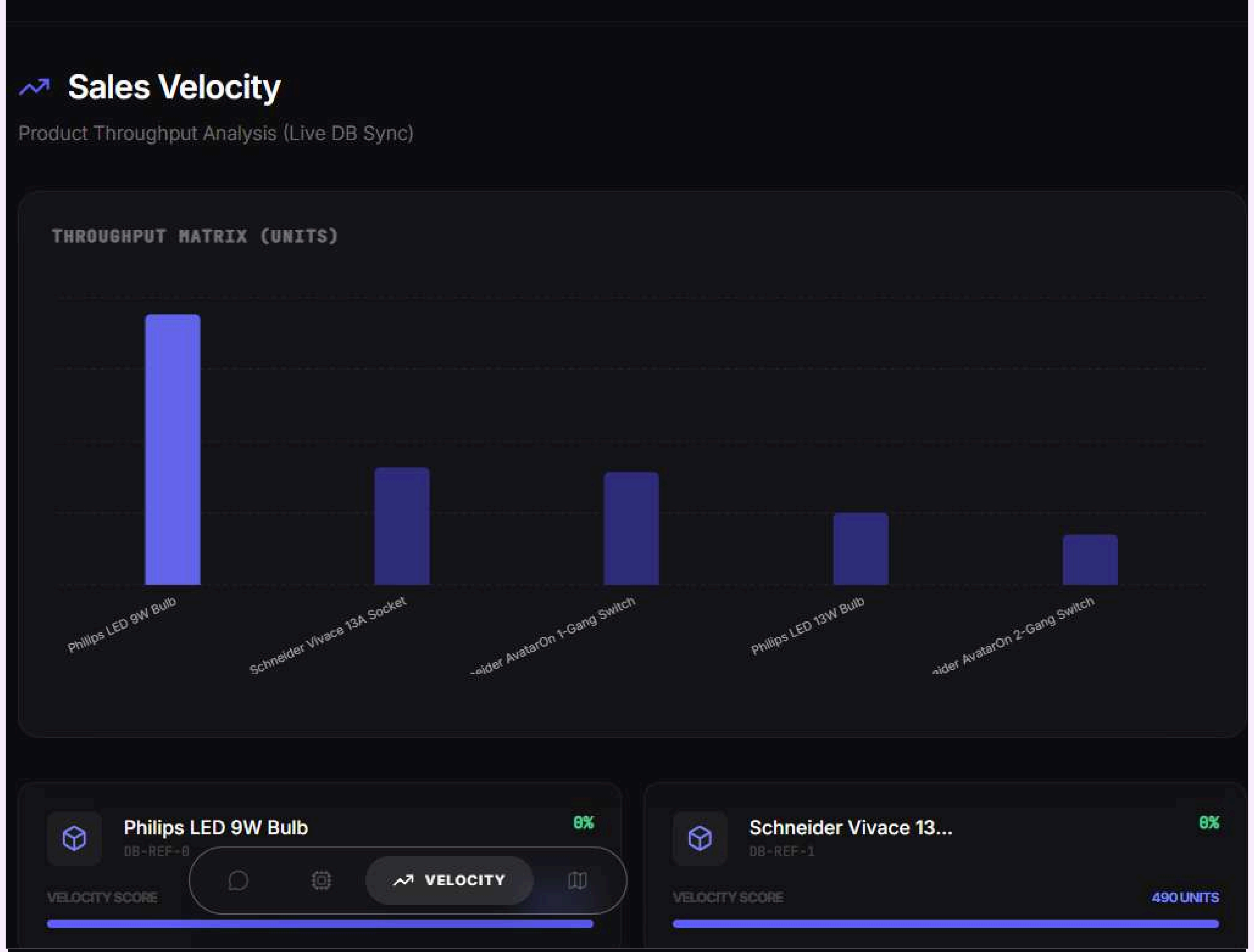
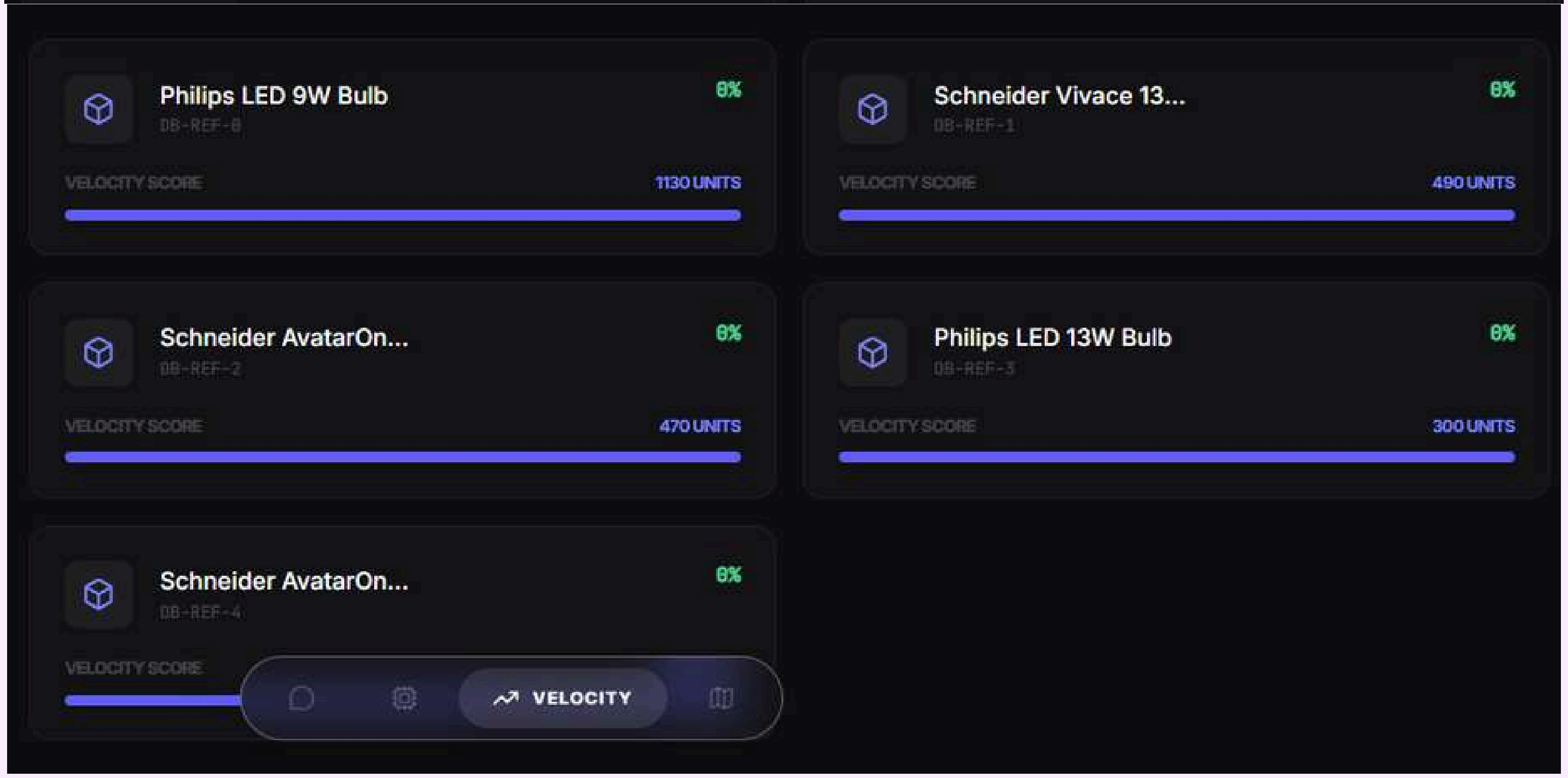
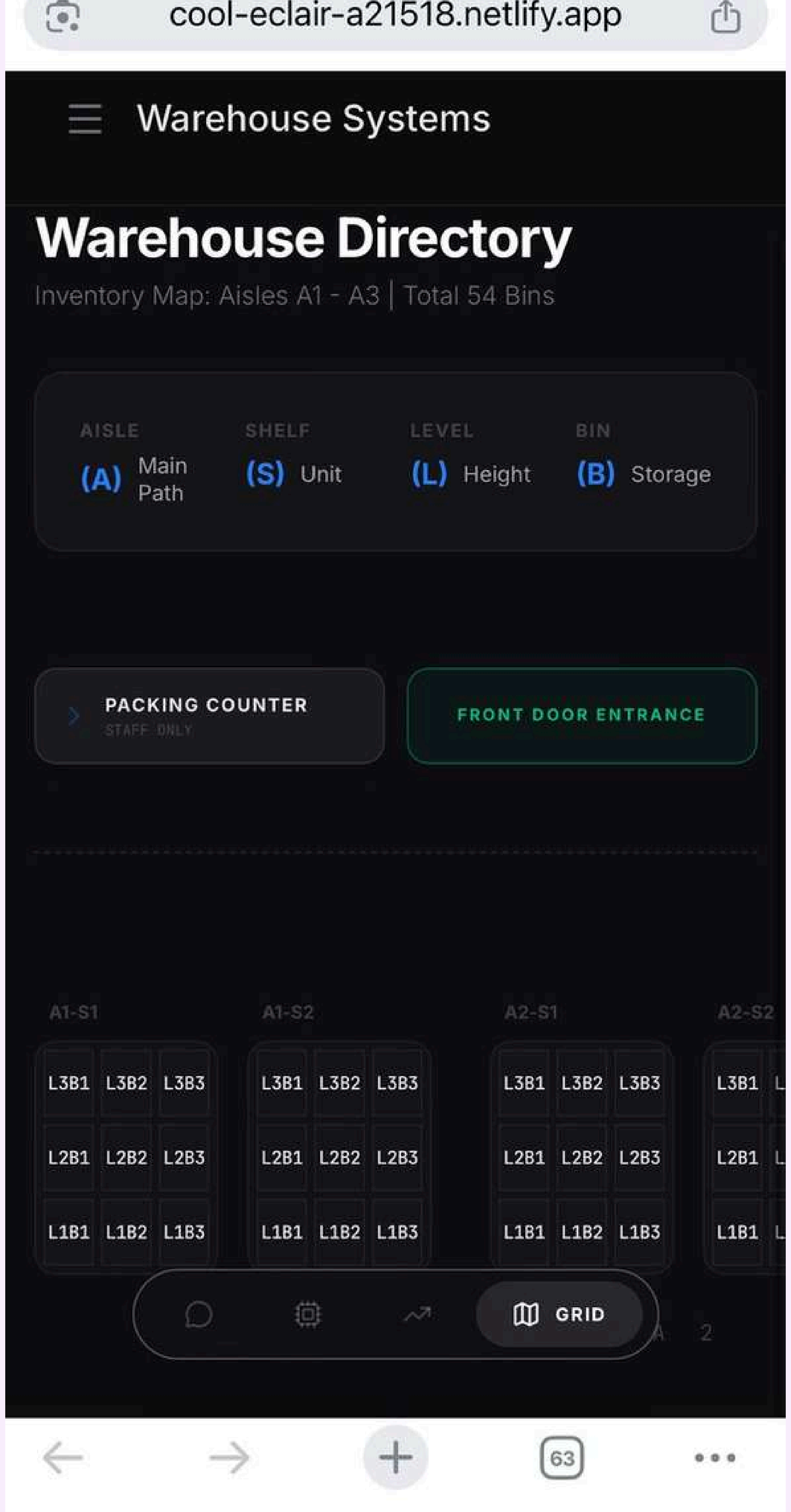
AISLE A2
0/18 active
Shelf A2-S1
REAL-TIME
L3B1
0%
L3B2
0%
L3B3
0%
L2B1
0%
L2B2
0%
L2B3
0%
L1B1
0%
L1B2
0%
L1B3
0%
Shelf A2-S2
REAL-TIME
CAPACITY
63


Sales Velocity
Product Throughput Analysis (Live DB Sync)
THROUGHPUT MATRIX (UNITS)
Philips LED 9W Bulb
Schneider Vivace 13A Socket
...eider AvatarOn 1-Gang Switch
Philips LED 13W Bulb
...eider AvatarOn 2-Gang Switch
Philips LED 9W Bulb
DB-REF-0
0%
VELOCITY SCORE
VELOCITY
Schneider Vivace 13...
DB-REF-1
0%
VELOCITY SCORE
490 UNITS


Philips LED 9W Bulb
DB-REF-0
0%
VELOCITY SCORE
1130 UNITS
Schneider Vivace 13...
DB-REF-1
0%
VELOCITY SCORE
490 UNITS
Schneider AvatarOn...
DB-REF-2
0%
VELOCITY SCORE
470 UNITS
Philips LED 13W Bulb
DB-REF-3
0%
VELOCITY SCORE
300 UNITS
Schneider AvatarOn...
DB-REF-4
0%
VELOCITY SCORE
VELOCITY


cool-eclair-a21518.netlify.app
Warehouse Systems
Warehouse Directory
Inventory Map: Aisles A1 - A3 | Total 54 Bins
AISLE
(A) Main Path
SHELF
(S) Unit
LEVEL
(L) Height
BIN
(B) Storage
PACKING COUNTER
STAFF ONLY
FRONT DOOR ENTRANCE
A1-S1
A1-S2
A2-S1
A2-S2
L3B1 L3B2 L3B3
L2B1 L2B2 L2B3
L1B1 L1B2 L1B3
GRID
63


CHAT FUNCTION

DATABASE

1. Bin database

	bin_id	status	blocked_...	max_wei...	max_volu...	accessibil...	Product1	Product1...	Product2	Product2...	capacity_...
1	A1-S1-L1-B1	Empty	Blocked	500	5	5	NULL	0	NULL	0	
2	A1-S1-L1-B2	Empty	Blocked	500	5	5	NULL	0	NULL	0	

2. Product database

	produ...	product_name	weight_kg	volume_...	product_description
1	K15VC	60 inch 3-Blade Ceiling Fan	6.5	0.045	KDK Regulator type ceiling fan, standard white finish, res...
2	K12V0	48 inch 3-Blade Compact Fan	5.2	0.035	KDK compact ceiling fan, standard white finish, ideal for ...

3. Sales database

	sale_id	sale_date	sale_time	produ...	quantity	customer_name
1	1	2026-04-15	08:15	PHL-9W-D	150	Maju Builders Sdn Bhd
2	2	2026-04-15	08:30	SCH-E8331	200	Maju Builders Sdn Bhd

MAIN TOOLS (JAVA METHOD)

1. Find Optimal Bin

   Find the best location for specific product based on the
   (i) quantity
   (ii) product weight,
   (iii) product volume,
   (iv)bin capacity,
   (v)blocked status
   (vi)sales velocity

2. Search Product by Descrition
   For new workers that dont know the product ID

3. Get product analysis
   Where the Agent can fetch usefull information from the
   product database

4. List warehouse inventory

5. Find product location

6. Get product dimension

7. Report accident

8. Notify all manager

9. Notify customer order arrival

10. Clear Aisle

Set Threshold: Item Speed = Required Accessibility Score (Max 5, Min 1).

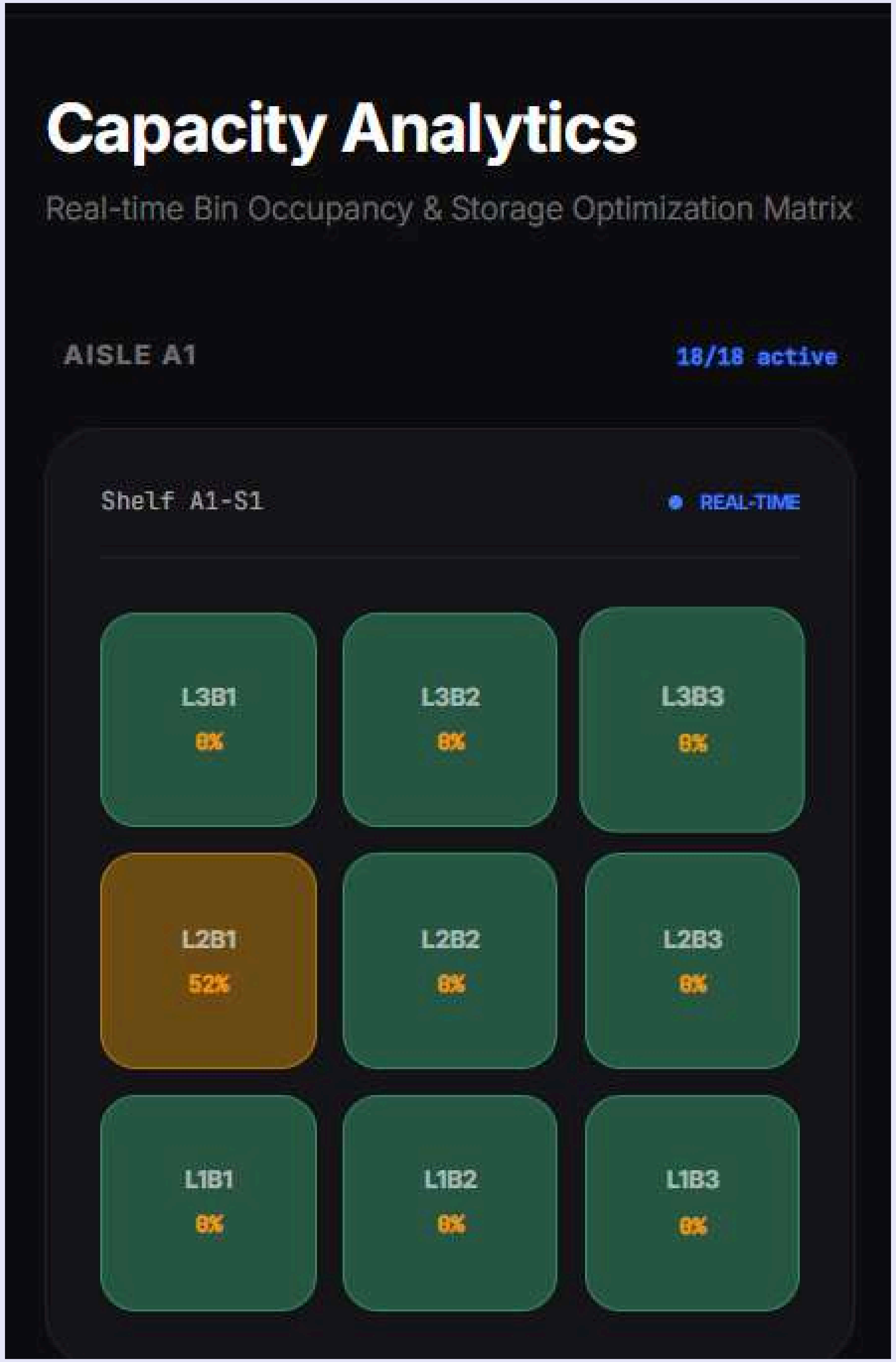
Filter Bins: Keep bins that are: 'Empty' or 'Half' , Unblocked , Not Excluded , and can fit ≥1 unit (&).

Rank Best Fit: Prioritize by: 'Half' status (Consolidate) > Same product inside (Affinity) > Lowest allowed score (Save premium bins).

Calculate Capacity: Max units placed = min(Weight capacity, Volume capacity).
Return Assignment .

HOW AI THINKS AT BACKEND

```
Worker: 5 KU50Y arrived here
[Zai is thinking...]

[SYSTEM] Zai activating skill: readLocalPurchaseOrder
[DATA RETRIEVED]
CustomerName,ProductID,Quantity,Priority
KL Rapid Construction,K15VC,40,High
Ah Chong Hardware,PHL-9W-D,150,Standard
City Office Renovations,K12V0,10,Standard

[Zai is analyzing the new data...]

[SYSTEM] Zai activating skill: getProductAnalysis
[DATA RETRIEVED]
Product: 20 inch Industrial Wall Fan, Weight: 8.50kg, Volume: 0.08m3, Velocity: 6
[Zai is analyzing the new data...]

[SYSTEM] Zai activating skill: findOptimalBin
[DATA RETRIEVED]
 A1-S1-L2-B1
[Zai is preparing final instruction...]

Zai: The remainder of the 5 KU50Y should be stored in bin  A1-S1-L2-B1. Please proceed to store the items and confirm when done.
```



USER INTERFACE


CUSTOMERS ORDER

HOW AI THINKS AT BACKEND

```
==============================================
   ZAI - WAREHOUSE INTELLIGENCE ONLINE
==============================================

Worker: A truck of 60 K15VC arrived
[SYSTEM] Arrival detected ? product: K15VC, quantity: 60
[Zai is thinking...]

[SYSTEM] Zai activating skill: readLocalCustomerOrder
[DATA RETRIEVED]
CustomerName,ProductID,Quantity,Priority
KL Rapid Construction,K15VC,40,High
Ah Chong Hardware,PHL-9W-D,150,Standard
City Office Renovations,K12V0,10,Standard


[SYSTEM CO CHECK]
[CO MATCH FOUND]
Product : K15VC
Customer: KL Rapid Construction
CO Qty  : 40 units
Priority: High
```

```
[SYSTEM] Zai activating skill: getProductAnalysis
[SYSTEM] AI resolved product ID: K15VC
[DATA RETRIEVED]
Product: 60 inch 3-Blade Ceiling Fan, Weight: 6.50kg, Volume: 0.05m3, Velocity: 47
[Zai is analyzing the new data...]

[SYSTEM] Zai activating skill: findOptimalBin
[SYSTEM] per-unit: K15VC => 6.5kg, 0.045m3, velocity=47, total qty=60
[DATA RETRIEVED]
Bin A1-S1-L2-B1: place 20 units.
[Zai is analyzing the new data...]
[RATE LIMIT] Attempt 1/3 ? waiting 5s...

Zai: Send 40 units to the Packing Counter for KL Rapid Construction (High priority).
Bin A1-S1-L2-B1: place 20 units.
Reply "done" when you have placed the items.
Worker:
```

```
ROUTING INSTRUCTION:
Route 40 units to the Packing Counter for KL Rapid Construction (High priority).
If arriving qty > 40, bin the remainder using getProductAnalysis + findOptimalBin.
If arriving qty <= 40, send ALL to Packing Counter ? do NOT bin anything.
```

Warehouse Alert Z.Ai
bot

✅ AISLE CLEARED
📍 **Location:** Aisle A1
🔓 **Action:** 0 bins unblocked. Zai has res

🚨 WAREHOUSE ALERT 🚨
📍 **Location:** Aisle A1
⚠️ **Issue:** accident
🔒 **Action:** 18 bins blocked. Zai is rerout

✅ AISLE CLEARED
📍 **Location:** Aisle A1
🔓 **Action:** 18 bins unblocked. Zai has re

📦 CUSTOMER ORDER ARRIVAL
🏷️ **Product:** PHL-9W-D
🏢 **Customer:** Ah Chong Hardware
🚚 **Arrived:** 60 units
⚡ **Priority:** Standard 3:52 PM

📦 CUSTOMER ORDER ARRIVAL
🏷️ **Product:** PHL-9W-D
🏢 **Customer:** Ah Chong Hardware
🚚 **Arrived:** 60 units
⚡ **Priority:** Standard 3:54 PM

📦 CUSTOMER ORDER ARRIVAL
🏷️ **Product:** K15VC
🏢 **Customer:** KL Rapid Construction
🚚 **Arrived:** 60 units
⚡ **Priority:** High 3:56 PM

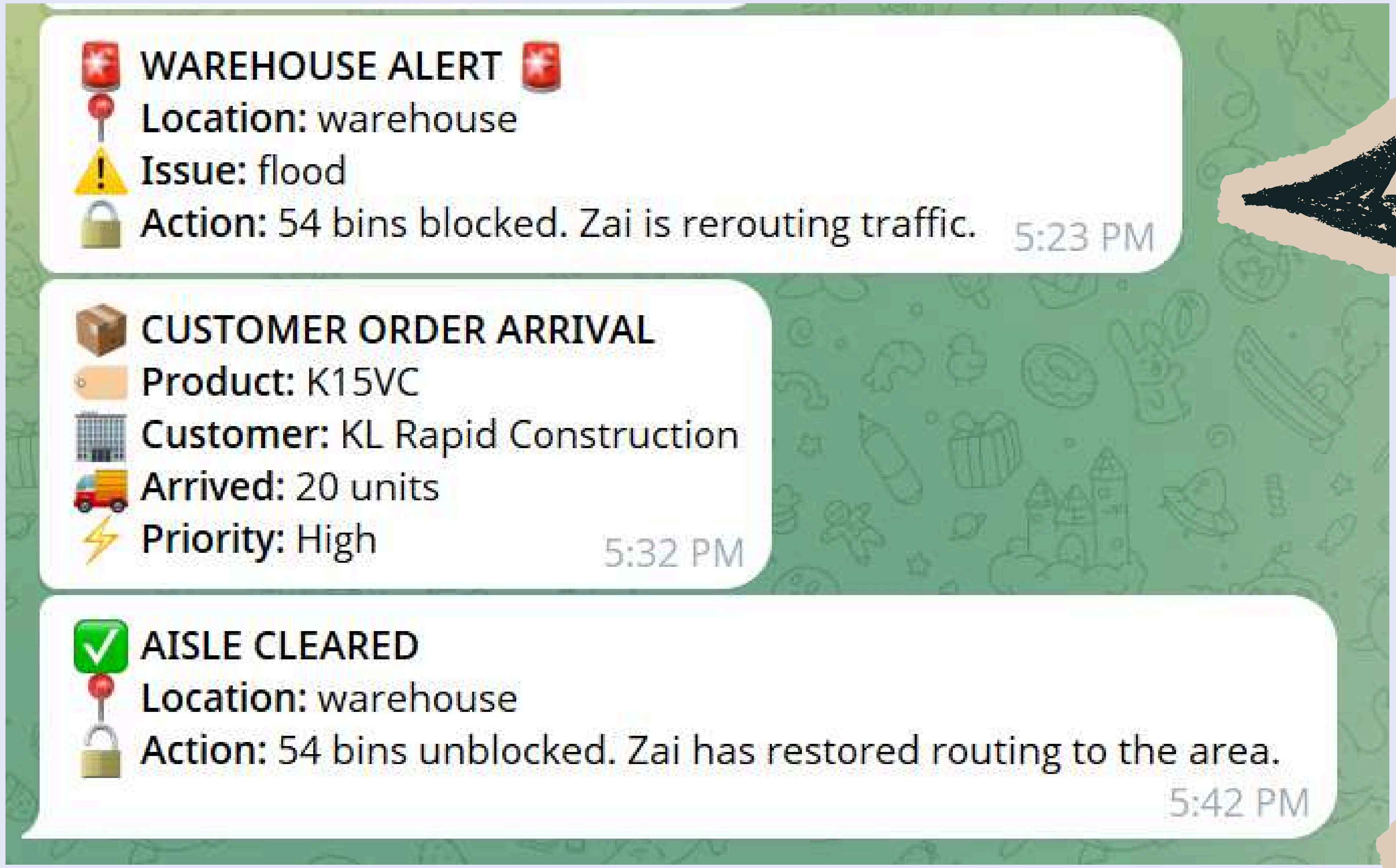
NOTIFY MESSAGE

ACCIDENT REPORT SYSTEM

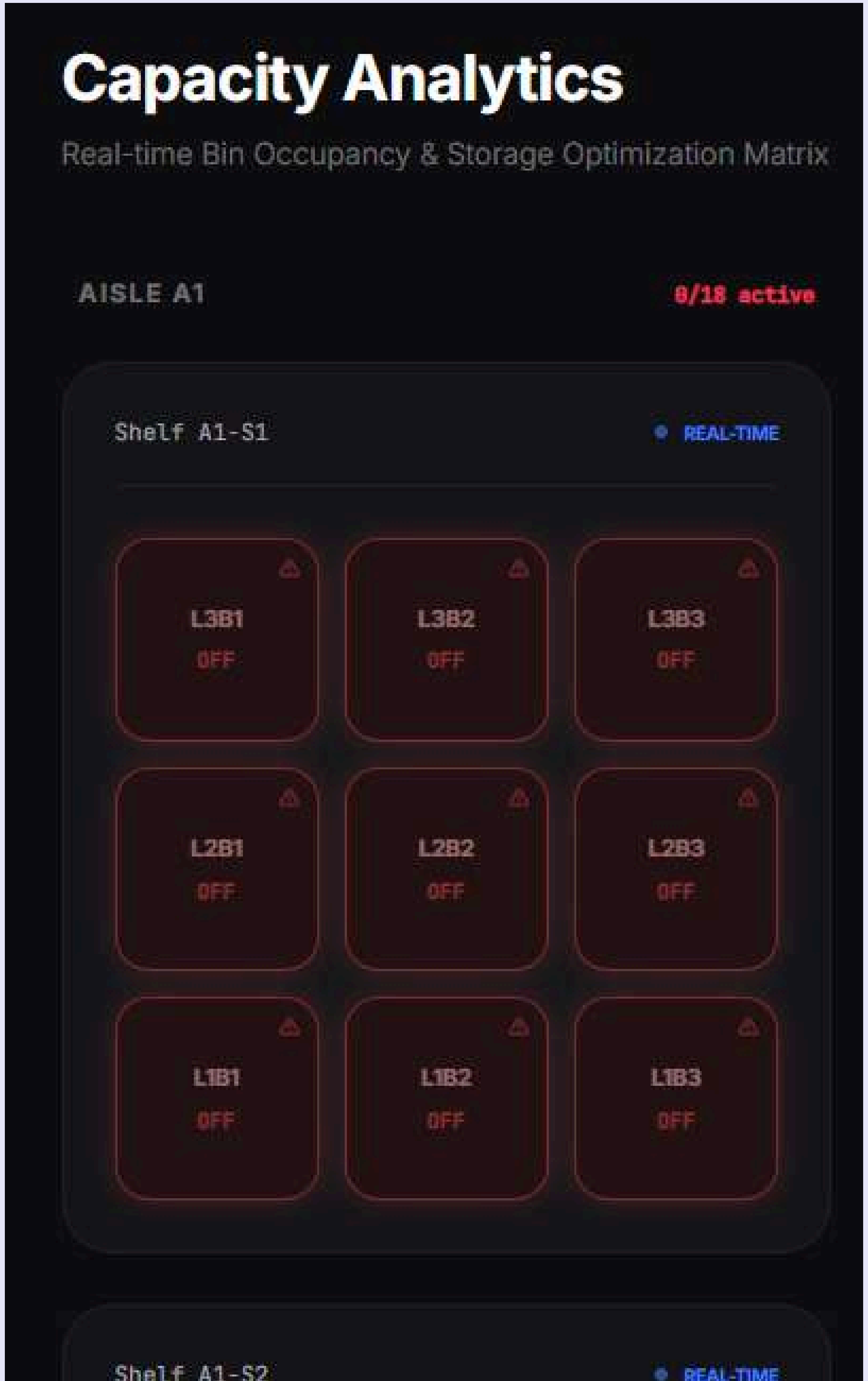
HOW AI THINKS AT BACKEND

```
Worker: flood happen
[Zai is thinking...]

[SYSTEM] Zai activating skill: reportAccident
[DB DEBUG] Blocking with pattern: %
[TELEGRAM -> 1324772413] Status: 200
[TELEGRAM -> 1324772413] Body:   {"ok":true,"r
[DATA RETRIEVED]
SUCCESS: Area 'warehouse' marked as blocked (54 bins locked). All managers notified via Telegram.
```

```
Worker: accident clear
[Zai is thinking...]

[SYSTEM] Zai activating skill: clearAisle
[DB DEBUG] Clearing with pattern: %
[TELEGRAM -> 1324772413] Status: 200
[TELEGRAM -> 1324772413] Body:   {"ok":tru
ATA RETRIEVED]
CCESS: Area 'warehouse' is now clear (54 bins unblocked). All managers notified via Telegra
```

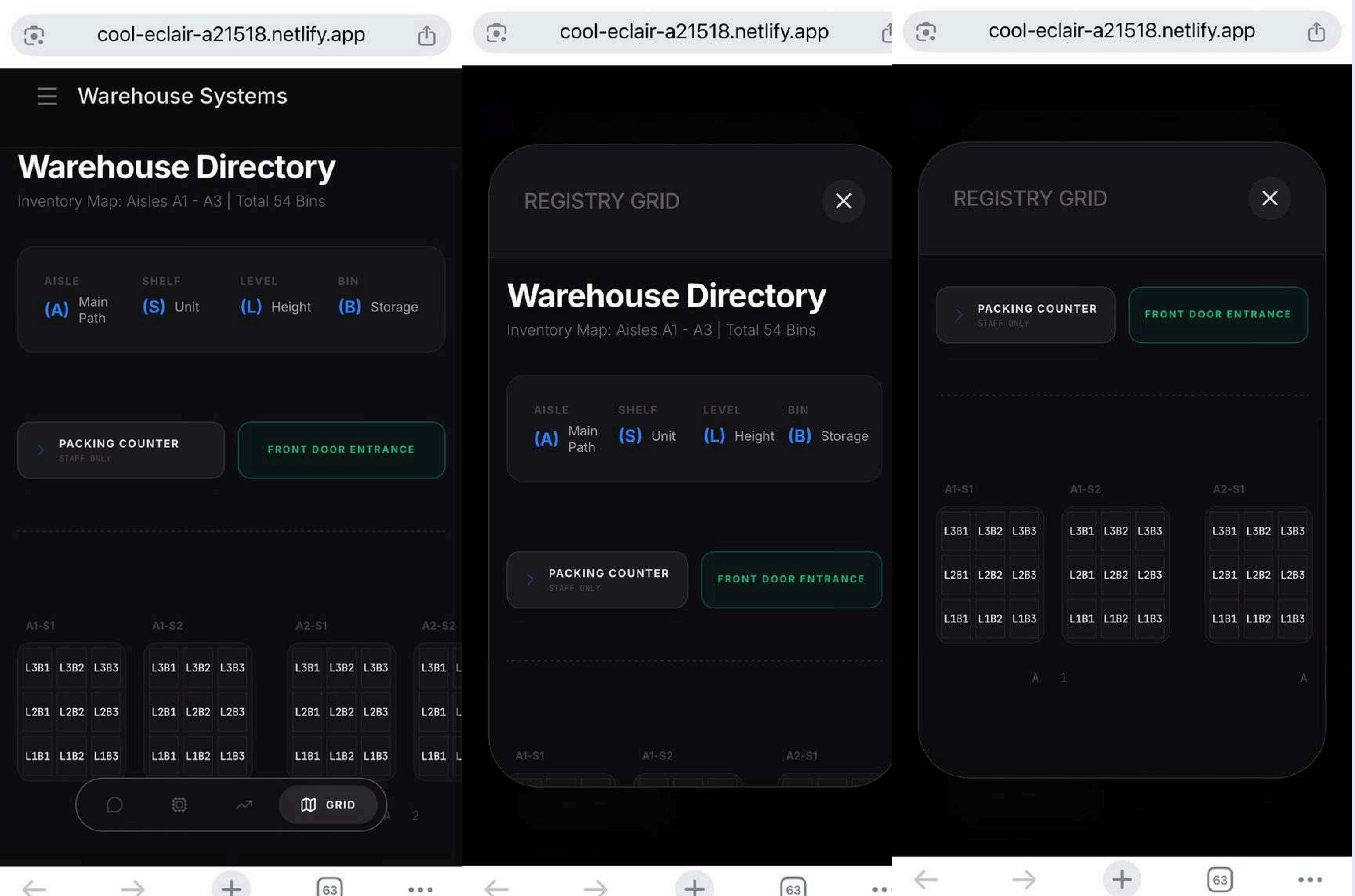


NOTIFY MESSAGE



USER INTERFACE


DIRECTORY MAP



(TAP) AND
(CHAT POPUP)

OVERALL BENEFITS

1. The "Intelligent Middleman" (Dynamic Cross-Docking)

- The system actively intercepts incoming deliveries against active customer purchase orders. It ensures workers never waste time binning a box that needs to be shipped out immediately.
- "Google Maps" for Inventory: Acts as an automated stock-finding system that provides instant product information and optimized routing.
- Maximizes Warehouse Space: Uses mathematical bin allocation to optimize spatial efficiency while strictly adhering to safety limits (e.g., weight capacity).

2. Operational Efficiency

- Focus on how the system empowers the everyday warehouse worker and removes bottlenecks.
- Eliminates Managerial Bottlenecks: Workers save massive amounts of time by interacting directly with their handheld devices rather than walking across the warehouse to track down a manager for the next task.
- Zero Miscommunication: Verbal instructions are easily misunderstood in a noisy warehouse (e.g., mishearing "twelve" as "five"). System-guided text instructions eliminate human error.
- * Persistent Task History: Workers never have to rely on memory; they can reference their instructions at any time, providing operational confidence and peace of mind.

OVERALL BENEFITS

3. Safety & System Reliability (Edge Cases)

- Real-Time Edge Case Rerouting: If an accident or spill happens in an aisle, workers can update the AI immediately. The system will intelligently route other workers away from the hazard to prevent secondary accidents or traffic jams.
- Deterministic AI, Zero Hallucinations: The system relies on strict database logic and mathematical guardrails rather than open-ended generation, ensuring the AI never hallucinates physical bin locations or inventory data.

THANK YOU!

TEOH JUN HONG

CHONG MING ZHE

ISAAC TOH

LEOW ZHEN XUN

DEMO !!

TEOH JUN HONG

CHONG MING ZHE

ISAAC TOH

LEOW ZHEN XUN